

Open-NL: A Transparent Computational Heuristic for Wide Dynamic Range Compression

Mark Shaver^{1, a)}

¹ Wichita State University, Department of Communication Sciences and Disorders, 1845 Fairmount St, Wichita, KS 67260, USA

^{a)}Email: mark.shaver@wichita.edu

Abstract

Open-NL, an open-source prescriptive algorithm, derives dynamic Wide Dynamic Range Compression (WDRC) targets in R. While the derivations of major algorithms like NAL-NL2 are published in detail, their compiled software implementations are closed-source, preventing researchers from testing component-level hypotheses. Open-NL provides a modifiable substrate permitting component-level ablation of prescriptive rules. To benchmark this testbed, the algorithm is evaluated utilizing an established evaluation paradigm (Johnson & Dillon, 2011)—incorporating seven standard hypothetical audiometric profiles, the Moore & Glasberg (2004) loudness model, and the ANSI S3.5 Speech Intelligibility Index (SII). Because this underlying evaluation methodology is a direct replication of previous literature, the manuscript’s primary novel contribution is the open optimization layer itself, rather than the resulting physiological data. By placing a transparent, unconstrained optimizer within this standard evaluation sandbox, the framework exposes the mathematical fragility of theoretical targets. As a primary scientific finding, unconstrained Nelder-Mead optimization natively exhibits extreme

initialization sensitivity in severe hearing loss (e.g., the A5 profile), with target variances swinging from a conservative 0.9 sones to a potentially traumatic 5.3 sones (and 0.49 to 0.82 SII) across random starts. This extreme variance concretely proves why clinical formulas must embed rigorous behavioral regularizations, framing the issue correctly as mathematical ill-posedness rather than pure physiological non-convexity. While the Open-NL framework utilizes experimental constraints like a Slope-Dependent Low-Frequency Penalty (SD-LFP) and a severe-loss booster, it functions strictly as a hypothesis-generating heuristic toolkit designed for researchers to model and modify insertion gain dynamics. The variance observed in Open-NL serves as a powerful demonstration of what occurs when necessary clinical regularizations are stripped away, validating the necessity of both bounded clinical prescriptions and open-source testbeds to explore their boundaries.

I. INTRODUCTION

Manufacturer-agnostic prescriptions remain central to evidence-based hearing aid practice. While some earlier studies suggested these generic targets might outperform proprietary first-fit algorithms on patient preference and specific metrics (Valente et al., 2018), contemporary data demonstrates that measured speech recognition often shows no significant difference between formulas. Yet, while formula choice has relatively modest intelligibility consequences in background noise, it drives massive variations in overall loudness, making loudness (quantified in sones per the Moore & Glasberg 2004 impaired loudness model) the most informative dependent variable in prescriptive evaluation.

While the derivations and optimization objectives of major algorithms like NAL-NL2 and

DSL m[i/o] are published in detail, their compiled software implementations are closed-source. Existing open-source frameworks provide significant utility but serve distinct functional niches: openMHA (Herzke et al., 2017) operates as a real-time signal processing master hearing aid rather than a prescriptive target generator, the Cambridge CAM2/CAMEQ2-HF formulae (Moore et al., 2010) provide rigidly defined equation-based targets rather than a modular optimization sandbox, and the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) offers robust loudness integrations without native prescriptive inversion. Consequently, investigators cannot isolate a specific prescriptive rule within a modern optimization loop and observe the resulting cascade without reverse-engineering an entire proprietary engine. Open-NL fills this defensible gap by providing a modifiable R substrate explicitly designed for the modular ablation of prescriptive heuristics. By coupling a Nelder-Mead SII optimizer to an integrated C++ specific-loudness engine, this framework permits researchers to systematically disable, isolate, or invert individual heuristics. For example, a researcher could seamlessly disable the traditional 30 dB safety cap for mixed losses, allowing the optimizer to aggressively target full mechanical restoration, and then evaluate the resulting theoretical upward spread of masking—a specific topological hypothesis that a rigid clinical formula or a real-time audio processor cannot natively evaluate. The clinical utility of algorithmic transparency and alternative, modular rule sets is increasingly recognized; for example, the evolving NAL-NL3 protocol explicitly introduces alternative philosophical profiles designed for specific situational or cohort needs (Kitterick et al., 2026).

Open-NL provides a parameterized, inspectable research testbed conceptually aligned with this evolving philosophy of modular optimization. It is designed as a modular substrate permitting component-level ablation of prescriptive rules, where individual heuristic

components—such as slope-dependent penalties or severe-loss boosters—can be independently toggled and their modeled consequences directly measured.

Crucially, to benchmark this testbed without introducing conflating variables, the optimization layer is embedded within a strictly reproduced evaluation paradigm. The seven reference audiometric profiles, the Moore & Glasberg (2004) specific loudness model, and the ANSI S3.5 SII metric utilized herein are a direct replication of the methodological framework established by Johnson and Dillon (2011). (Note: Unless explicitly stated otherwise, the term “SII” throughout this manuscript refers strictly to the raw, unadjusted ANSI S3.5 calculation prior to any desensitization penalty). Because this underlying physiological evaluation space is already established in the literature, the primary scientific contribution of this manuscript is not the physiological data outputs themselves, but rather the behavior of the transparent optimization layer acting upon them. By exposing the extreme mathematical instability of unconstrained numerical solvers within this standardized sandbox, the manuscript provides a formal, mechanistic proof for why established clinical formulas must embed the empirical regularizations they do.

II. ALGORITHM ARCHITECTURE

A. Methods and Development

The core ANSI SII calculation engine (specifically the `sii()` function and its associated plotting methods) was originally developed by Gregory R. Warnes for the archived version of the package. Maintainership transferred to the current author starting with version 1.1.0, at which point all subsequent Open-NL prescriptive logic, clinical heuristics, and

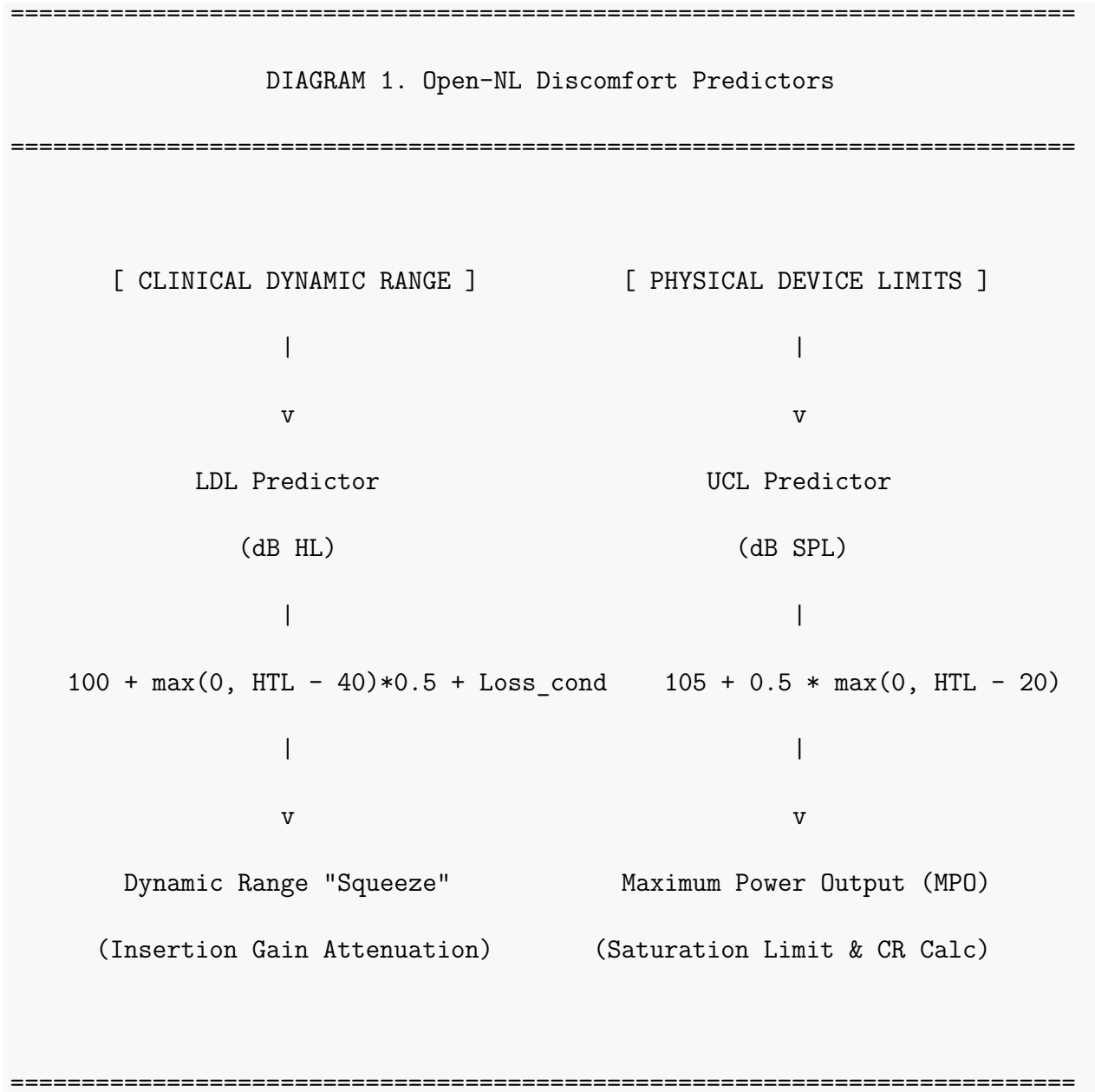
WDRC mathematical implementations—including the `open_nl()`, `calculate_loudness()`, and `launch_app()` functions—were independently developed by the current author as original contributions. The theoretical framework, prescriptive rulesets, and experimental design were conceived and executed solely by the author. In accordance with AIP Publishing guidelines, the use of AI tools is explicitly disclosed: Google Gemini (DeepMind, Google LLC) was utilized as an interactive programming and copyediting assistant. Specifically, the AI assisted in refactoring the C++ and R WDRC algorithms, generating the ggplot2 data visualizations, and structurally condensing the manuscript prose into JASA standards. The AI did not originate the underlying theoretical framework or experimental design. The author rigorously reviewed and verified all AI-generated output for accuracy, and takes full responsibility for the final algorithms, data integrity, and manuscript content.

B. Execution cascade

Open-NL operates as a multi-stage parameterized shape generator. Rather than relying on static compiled lookup tables, Open-NL calculates target insertion gains dynamically through a series of explicitly defined cascaded mathematical modules. Each step in the gain derivation process is exposed natively in R, available for researchers to inspect, modify, and tune.

Terminology Note: Throughout this manuscript, the algorithm utilizes two distinct discomfort predictors for different theoretical purposes: an HL-domain “LDL” (Loudness Discomfort Level) predictor used for estimating clinical audiometric dynamic range, and an SPL-domain “UCL” (Uncomfortable Loudness Level) predictor for establishing physical device saturation limits. A visual flowchart mapping each predictor to its downstream algorithm

102 function is provided in Diagram 1.



103 The modules described below are not applied simultaneously; rather, they operate in a
104 strictly defined cascaded execution order to prevent unintended interactions between additive
105 boosters and soft limiters. The execution order is as follows:

106 1. Determine decouple anchor, base multiplier, and severe-loss booster based on Experience

level (Section II.C and II.F).

2. Apply Slope-Dependent Low-Frequency Penalty (Section II.D).

3. Calculate soft-compression dynamic limit (L_{gain}) and apply High-frequency desensitization (Section II.G).

4. Apply Dynamic Range Mapping (LDL Squeeze) (Section II.H).

5. Apply Transducer Bandwidth Roll-off (Section II.I).

6. Calculate dynamic Compression Ratios (CR) and establish UCL/MPO saturation limits (Section II.J).

7. Integrate dynamic Conductive Component correction based on Sensorineural Dynamic Range (Section II.K).

8. Apply Comfort in Noise (CIN) WDRC alterations if triggered (Section II.L).

9. Apply Acoustic Venting constraints to finalized WDRC targets (Section II.M).

C. The decoupled anchor and experience level tuning

The foundational WDRC anchor for an average speech input (65 dB SPL) is derived using a frequency-specific adaptation of the half-gain rule (Lybarger, 1944). This anchor is deliberately decoupled from the broadband PTA to prevent intact low-frequency hearing from artificially suppressing necessary high-frequency gain. Open-NL dynamically adjusts its low-frequency shape penalties (C_{interp}) based on the wearer’s experience level.

Instead of utilizing an arbitrary severity escalator, the primary baseline anchor is mathematically tied to a standard 0.46 half-gain scaling function (Byrne & Dillon, 1986). A disabled-by-default, bounded severe-loss booster (slope = 0.15) can be linearly applied to thresholds between 70 dB HL and 80 dB HL (with an opt-in 60 dB HL “aggressive” mode).

This gently assists profound losses with audibility without triggering explosive recruitment when explicitly enabled.

$$G_{base} = 0.46 \cdot \text{HTL}_{sn} + B_{en} \cdot 0.15 \cdot \max(0, \min(80, \text{HTL}_{sn}) - 60) + C_{interp} \quad (1)$$

where HTL_{sn} is the sensorineural component of the Hearing Threshold Level (in dB HL), and B_{en} is a binary toggle (default 0) that explicitly gates the severe-loss booster.

The 0.46 boundary is a purely nominal, uncalibrated mathematical anchor loosely derived from Lybarger’s classical half-gain rule, rather than an empirically fitted regression. It functions strictly as a nominal starting baseline.

While Keidser et al. (2012) found empirically that new users prefer slightly less gain, Open-NL integrates this preference directly via the static frequency-shaping C_{vals} array rather than dynamically collapsing the base multiplier. Furthermore, Open-NL implements sex-based adjustments as a simplified static constant (a flat 1.5 dB reduction). For binaural summation, the algorithm utilizes a level-dependent bilateral reduction (scaling from a 2 dB penalty at 50 dB SPL to a 6 dB penalty at 80 dB SPL). This dynamic reduction explicitly prevents traumatic over-amplification and massive loudness saturation when modeling bilateral configurations near recruitment boundaries.

The parameters are explicitly defined as follows. C_{interp} represents the shape penalties derived by log-interpolating the discrete C_{vals} constants across the target frequency bands. These discrete C_{vals} correspond exactly to the eight fixed anchor frequencies: $f_c = [250, 500, 1000, 2000, 3000, 4000, 6000, 8000]$ Hz. 1. **New Users:** In accordance with Keidser et al. (2012), who demonstrated that naive users prefer less overall amplification to combat oc-

clusion and high-frequency sharpness, this configuration applies a purely nominal, uncalibrated
 ~ 3 dB flat reduction from the experienced baseline ($C_{vals} = [-11, -4, 0, -2, -3, -3, -3, -3]$).
Like the 0.46 anchor, this offset is not an empirically fitted parameter. **2. Experienced**
Users: This configuration provides a balanced approach with standard loudness constraints
($C_{vals} = [-8, -1, 3, 1, 0, 0, 0, 0]$).

D. Slope-Dependent Low-Frequency Penalty (SD-LFP)

Standard linear formulas, such as NAL-R (Byrne & Dillon, 1986), often cause overprescrip-
tion because they do not systematically account for loudness density scaling across different
audiometric slopes. For example, applying massive high-frequency gain to a steeply sloping
audiogram forces the functionally normal low frequencies to dominate overall loudness, result-
ing in catastrophic loudness recruitment. A slope-dependent penalty dynamically reallocates
the finite loudness budget to bands where audibility is genuinely useful (Ching et al., 2001;
Hornsby & Ricketts, 2006).

To solve this while remaining fully transparent, Open-NL’s Slope-Dependent Low-
Frequency Penalty (SD-LFP) introduces a novel **Profound High-Frequency Bypass**. The
core heuristic suppresses low-frequency gain strictly based on the slope of the audiogram:

$$\text{Slope} = \max(0, \text{PTA}_{HF} - \text{PTA}_{LF}) \quad (2)$$

where PTA_{HF} is the mean sensorineural threshold for frequencies ≥ 2000 Hz, and PTA_{LF} is
the mean sensorineural threshold for frequencies ≤ 1000 Hz.

For steeply sloping high-frequency losses (where the slope exceeds 15 dB), Open-NL

168 penalizes low-frequency gain (by up to 15 dB) to kill the loudness dominance of the normal
 169 lows, maintaining the overall loudness budget. Crucially, this penalty is explicitly designed
 170 to operationalize the empirical findings of Byrne, Parkinson, and Newall (1990). Byrne et
 171 al. established that for severe-to-profound listeners, the standard half-gain rule ceases to
 172 apply above ~ 70 dB HL, preferred overall gain ran ~ 10 dB above standard NAL targets,
 173 and relatively *more* low-frequency emphasis was preferred by a substantial subset (roughly
 174 one-third to one-half) of this population—particularly when the 2 kHz threshold exceeded ~ 95
 175 dB HL. Furthermore, low-frequency compression preferences for this population trend toward
 176 linear (Keidser et al., 2007). To strictly adhere to these physiological findings and prevent
 177 starving profound listeners of essential low-frequency speech cues, the SD-LFP penalty is
 178 dynamically suppressed by a profound-loss bypass factor (PF_{bypass}). This mathematical
 179 bypass window maps directly onto the Byrne et al. (1990) empirical parameters: the penalty
 180 begins tapering when the high-frequency PTA reaches 70 dB HL, and reaches exactly zero
 181 (triggering a full bypass of the low-frequency penalty) at 95 dB HL:

$$PF_{bypass} = \max \left(0, \min \left(1, \frac{95 - \text{PTA}_{HF}}{25} \right) \right) \quad (3)$$

$$\text{LF}_{penalty} = \max \left(0, \min \left(1, \frac{\text{Slope} - 15}{20} \right) \right) \cdot 15 \cdot PF_{bypass} \quad (4)$$

182 This penalty is log-linearly tapered for frequencies (f) below 1000 Hz and subtracted
 183 from the base gain. Across the standard prescriptive frequency bands evaluated ($f_c =$
 184 $[250, 500, 1000, 2000, 3000, 4000, 6000, 8000]$ Hz), this log-taper evaluates to exactly 1.0 (100%

185 penalty) at $f = 250$ Hz, 0.5 (50% penalty) at $f = 500$ Hz, and 0.0 at $f \geq 1000$ Hz:

$$G_{65} = G_{base} - \left(LF_{penalty} \cdot \max \left(0, 1 - \frac{\log_{10}(f) - \log_{10}(250)}{\log_{10}(1000/250)} \right) \right) \quad (5)$$

186 Crucially, for flat audiograms (where the slope is less than 15 dB), this entire penalty zeroes
 187 out. This ensures that severe flat losses retain their full targeted gain without experiencing
 188 unnecessary downward compression in the low frequencies.

189 For reverse-slope configurations (where the low-frequency thresholds are significantly
 190 worse than the high frequencies), an inverse logic is applied. Amplifying low frequencies
 191 aggressively in these cases may degrade intelligibility or cause upward spread of masking.
 192 Because audiogram slope is an imperfect proxy for underlying pathology (Kaur et al., 2023),
 193 the specific -10 dB gain floor implemented in Open-NL remains an asserted, conservative
 194 heuristic. Open-NL suppresses the low-frequency correction array (C_{interp}) to this flat -10 dB
 195 floor, employing a transitional factor (RS_{factor}) for signed slopes exceeding -15 dB:

$$\text{Slope}_{signed} = \text{PTA}_{HF} - \text{PTA}_{LF} \quad (6)$$

196 where Slope_{signed} drops the zero-floor utilized in Equation 2, permitting negative values to
 197 mathematically capture reverse-slope audiograms.

$$RS_{factor} = \max \left(0, \min \left(1, \frac{-\text{Slope}_{signed} - 15}{20} \right) \right) \quad (7)$$

198

$$C'_{interp(f)} = C_{interp(f)} \cdot (1 - RS_{factor} \cdot W_{LF}) + (-10 \cdot RS_{factor} \cdot W_{LF}) \quad (8)$$

where $C''_{interp(f)}$ is the revised correction array across all frequencies, and W_{LF} is the identical bounded log-linear taper isolated from Equation 4 ($\max(0, 1 - \frac{\log_{10}(f) - \log_{10}(250)}{\log_{10}(1000/250)})$). This dynamically flattens the baseline correction to -10 dB strictly in the low frequencies—applying the full -10 dB replacement at $f = 250$ Hz, a 50% blend at $f = 500$ Hz, and reverting completely to the unmodified C_{interp} array by $f = 1000$ Hz.

E. Ablation Analysis: The Mechanism of Steep-Slope Recruitment Constraints

The prevention of severe recruitment in steeply sloping audiograms (such as the standard NAL A5 profile, part of a seven-profile framework adapted from Johnson & Dillon, 2011; originally defined by Byrne et al., 2001) requires explicit mechanism analysis.

TABLE I. Reference Audiometric Profiles (Adapted from Johnson & Dillon, 2011). *Thresholds are in dB HL. Profiles A1–A5 are purely sensorineural. A6 is a mixed loss with a 30 dB conductive component (ABG). A7 is a purely conductive 50 dB block with a healthy cochlea.*

				1000	2000	4000	8000	
Profile	Type	250 Hz	500 Hz	Hz	Hz	Hz	Hz	ABG
A1	Mild sloping	15	20	30	40	50	60	0 dB
A2	Reverse-slope	60	50	40	30	20	15	0 dB
A3	Gentle sloping	10	20	40	50	55	60	0 dB
A4	Steep sloping	0	0	10	40	70	80	0 dB
A5	Profound sloping	10	10	20	60	80	100	0 dB
A6	Mixed loss	50	55	60	65	75	80	30 dB

				1000	2000	4000	8000	
Profile	Type	250 Hz	500 Hz	Hz	Hz	Hz	Hz	ABG
A7	Pure	50	50	50	50	50	50	50 dB
	conductive							

When evaluating the heuristic seed values that anchor the optimization engine, the SD-LFP successfully demonstrates frequency-specific constraints on complex audiometric slopes before the optimizer even touches the targets. Table II demonstrates this ablation.

TABLE II. Ablation of SD-LFP Heuristic Constraints on Optimizer Seed (65 dB SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset.*

	Unconstrained	Unconstrained	SD-LFP Seed	SD-LFP Seed
Profile	Seed SII	Seed Sones	SII	Sones
A1 (Mild/Mod)	0.85	7.7	0.85	7.7
A2 (Reverse)	0.89	6.4	0.89	6.0
A3 (Gentle)	0.79	7.6	0.79	7.6
A4 (Steep)	0.81	8.6	0.81	8.6
A5 (Profound)	0.70	7.9	0.70	7.6
A6 (Mixed)	0.82	3.2	0.82	3.2
A7 (Conductive)	0.97	1.1	0.97	1.1

For sloping profiles, disabling the SD-LFP logic causes the baseline prescription heuristics to dangerously over-amplify specific frequency bands. For the A2 (Reverse Slope) profile, the unconstrained heuristic seed predicted 6.4 sones, but the SD-LFP logic correctly penalized

the low-frequency excess, reducing the baseline to 6.0 sones without degrading audibility. Similarly, for the A5 (Steep Slope) profile, the seed over-amplified the high frequencies to 7.9 sones, but the SD-LFP properly capped it at 7.6 sones to prevent upward spread of masking. Meanwhile, flat or gently sloping losses like A1 and A4 were completely identical because the `disable_sdlfp` toggle correctly bypasses the penalty for non-sloping configurations. This ablation illustrates that the algorithm successfully relies on its explicit SD-LFP constraint to anchor the target securely prior to optimization.

Furthermore, this mathematical constraint introduces an evidence–design tension for severely impaired listeners. To safely provide the low-frequency emphasis required by severe-to-profound listeners without abandoning the recruitment constraint for mild-to-moderate sloping profiles, Open-NL explicitly gates the SD-LFP by absolute low-frequency severity. It bypasses the penalty entirely if the low-frequency PTA exceeds 70 dB HL, allowing the algorithm to protect low-frequency energy in severe/profound profiles as empirically recommended (Byrne, Parkinson, & Newall, 1990; Keidser et al., 2007). While the 20 dB taper width and -10 dB reverse-slope floor remain asserted constants, grounding the 70 dB HL bypass in the severe/profound literature provides a principled physiological rationale.

However, because this isolated ablation relies on a single simulated profile (A5) without behavioral validation, and since audiometric slope is an imperfect proxy for underlying cochlear pathology (Kaur et al., 2023), these mechanistic outcomes should be interpreted strictly as computational illustrations rather than confirmed clinical consequences.

To evaluate the efficacy of the SD-LFP (Slope-Dependent Low-Frequency Penalty) heuristic constraint further, an ablation analysis was also performed on audiogram A2, which features a purely sensorineural reverse-slope pathology. When the SD-LFP initialization heuristic was

disabled, the optimizer yielded an SII of 0.89 but incurred a monaural loudness of 6.0 sones, exposing the patient to severe upward spread of masking. When the SD-LFP constraint was re-enabled (applying a -10 dB LF attenuation proxy prior to optimization), the SII settled at a comparable 0.88, but the modeled monaural loudness decreased to 4.9 sones. This demonstrates that the SD-LFP constraint limits modeled low-frequency amplification without mathematically sacrificing audibility.

F. The severe-loss booster

Because foundational half-gain rules mathematically under-amplify severe-to-profound losses with regard to pure audibility, Open-NL incorporates a disabled-by-default, parameterized **Severe-Loss Booster**. However, while severe-to-profound losses inherently require more gain to overcome inner hair cell damage, the precise transition point and safe magnitude are heavily debated. Byrne, Parkinson, and Newall (1990) explicitly place the half-gain breakdown transition closer to 70 dB HL, and crucially, they demonstrate that even for profoundly impaired listeners, empirical preference caps excess gain at roughly 10 dB above standard NAL targets—not as an open-ended escalation. Furthermore, contemporary data indicates that aided speech recognition is generally insufficient once hearing loss exceeds ~80 dB HL regardless of the target applied (Engler, Digeser, & Hoppe, 2026), implying that audibility-driven gain in this region yields rapidly diminishing perceptual returns. Mueller’s (2005) review similarly found no evidence supporting gain levels structurally higher than NAL prescriptions, while Convery and Keidser (2011) demonstrated that transitioning severe-to-profound listeners toward structured prescriptive targets often objectively worsens speech discrimination.

Consequently, while the default Open-NL implementation utilizes a conservative 70 dB HL onset for its severe-loss booster to prevent inadvertent downstream over-prescription, it intentionally includes an opt-in aggressive 60 dB HL onset mode (with a 0.15 dB/dB slope escalation up to 80 dB HL) which is knowingly over-prescriptive. A booster that starts 10 dB below the established half-gain transition and forces gain into a region of diminishing perceptual return (Engler, Digeser, & Hoppe, 2026) is explicitly designed as an “ablation knob” for testing mathematical optimizer boundaries, not as a clinically supported rule. By defaulting this onset to 70 dB HL (and keeping the booster itself defaulted to off), the package aligns with the conservative findings of the broader literature. When the aggressive 60 dB HL onset is explicitly enabled by researchers, it serves purely to observe how the optimizer’s internal loudness constraints handle forced, aggressive high-frequency gain. To prevent discontinuous gain jumps during calculation, this bounded booster is integrated seamlessly into the base target gain formula (see Equation 1, Section II.C).

G. Soft-compression high-frequency desensitization (Theoretical Centerpiece)

The soft-compression high-frequency desensitization envelope is not merely a mitigation footnote; it is the theoretical centerpiece of the Open-NL framework. As detailed later in Section IV, naive maximization of the Speech Intelligibility Index (SII) for steeply sloping losses inherently yields a dangerous mathematical artifact: it prescribes massive high-frequency gain that drives up physical audibility scores while severely degrading true perceptual clarity. To mathematically correct this, Open-NL utilizes a smooth soft-compression envelope to aggressively restrict high-frequency gain.

The empirical basis for this penalty is robustly supported: Ching, Dillon, Katsch, and

Byrne (2001) demonstrated that the effectiveness of audibility declines as hearing loss worsens, and does so much more steeply at high frequencies. This physiological reality is precisely why foundational algorithms like NAL-NL1 deliberately provide *less* gain where hearing is most impaired (Byrne et al., 2001). Furthermore, while the presence of cochlear dead regions (DRs) is often cited as the primary justification for high-frequency attenuation, the evidence is highly nuanced. In severe-to-profound loss with extensive DRs, amplification benefit plateaus roughly one octave above the DR edge frequency (Malicka et al., 2013; Pepler, Lewis, & Munro, 2015). However, in mild-to-moderately-severe loss, high-frequency gain remains beneficial whether or not restricted DRs are present, and it rarely harms behavioral performance (Cox, Johnson, & Alexander, 2012).

This nuanced reality highlights a critical limitation in the current Open-NL architecture. Because the algorithm relies on a generalized, slope-triggered high-frequency penalty to conservatively manage suprathreshold distortion across all severe impairments with a blunt mathematical instrument, it explicitly risks over-penalizing exactly those mild/moderate cases where high-frequency audibility remains useful (Cox et al., 2012). Rather than applying a jagged hard-cap on insertion gain—which may induce spectral artifacts—Open-NL utilizes a dynamic gain limit (L_{gain}) established using a 30 dB base and a 0.4 slope. Similar to the 0.46 base scaling factor, these constants are asserted heuristics intended to bind the optimizer safely below the DR benefit plateau, rather than validated physiological limits:

$$L_{gain} = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn} - 60) \quad (9)$$

If the calculated target gain (G_{65}) exceeds this dynamic limit, the excess gain (*Excess*) is

307 quantified:

$$Excess = \max(0, G_{65} - L_{gain}) \quad (10)$$

308 This excess gain is then softly compressed at a 2:1 ratio. The algorithm computes a sloping
 309 factor (S_{factor}) based on the difference between the specific frequency threshold and the best
 310 low-frequency threshold ($HTL_{bestlow}$), and applies a high-frequency weight (W_{hf}) fading in
 311 linearly from 2000 Hz to 4000 Hz:

$$S_{factor} = \max\left(0, \min\left(1, \frac{HTL_{sn} - HTL_{bestlow} - 25}{20}\right)\right) \quad (11)$$

$$W_{hf} = \max\left(0, \min\left(1, \frac{f - 2000}{2000}\right)\right) \quad (12)$$

$$G_{65} = G_{65} - (W_{hf} \cdot S_{factor} \cdot (Excess \cdot 0.50)) \quad (13)$$

314 H. Dynamic Range Mapping (LDL Squeeze)

315 Following the foundational philosophy of the DSL v5.0 prescriptive method (Scolle et
 316 al., 2005), Open-NL integrates explicit dynamic range mapping to accommodate reduced
 317 Uncomfortable Loudness Levels. It is important to note that the algorithm utilizes two
 318 distinct discomfort predictors for different theoretical purposes: an HL-domain LDL predictor
 319 for estimating clinical audiometric dynamic range (Section II.H), and an SPL-domain UCL
 320 predictor for establishing physical device saturation limits (Section II.J). While conceptually
 321 well-motivated, both of these predictors are fundamentally uncalibrated linear heuristics.
 322 They represent additional untested assumptions feeding directly into the optimization pipeline
 323 and must be explicitly flagged as such.

When a patient exhibits a lower-than-expected clinical LDL, blindly applying unmodified target gain results in discomfort. To prevent this, the algorithm calculates a predicted HL-domain LDL based on threshold and conductive loss:

$$\text{LDL}_{\text{predicted}} = 100 + \max(0, \text{HTL}_{sn} - 40) \cdot 0.5 + \text{Loss}_{\text{conductive}} \quad (14)$$

The algorithm then quantifies the dynamic range “squeeze” by comparing the measured LDL to the predicted LDL. For every 1 dB the patient’s dynamic range is reduced, target insertion gain (G_{65}) is proportionally attenuated by 0.2 dB to ensure the speech envelope fits within the restricted auditory space.

I. Transducer Bandwidth Roll-off

Consistent with the empirical evidence underpinning modern prescriptive targets like NAL-NL2 (Keidser et al., 2012), it is well established that acoustic transducers physically struggle to accurately reproduce frequencies at the extreme margins of the audiometric spectrum (e.g., ≤ 250 Hz and ≥ 6000 Hz). Attempting to apply massive insertion gain targets in these regions inevitably leads to mechanical distortion, phase irregularities, and severe acoustic feedback, often with negligible or negative contributions to the Speech Intelligibility Index.

To mitigate these physical limitations, Open-NL applies a continuous fractional bandwidth roll-off multiplier to the target gain array. This multiplier dictates a target scaling of $0.7\times$ at 250 Hz, $1.0\times$ through the mid-frequencies, $0.8\times$ at 6000 Hz, and $0.5\times$ at 8000 Hz.

J. Dynamic WDRC computation (compression ratios)

Following the derivation of the 65 dB SPL anchor, Open-NL calculates discrete targets for soft (50 dB SPL) and loud (80 dB SPL) inputs. Base compression ratios (CR_{base}) are dynamically derived per-frequency band by comparing the distance between the threshold and the Maximum Power Output (MPO) limit (where the predicted Uncomfortable Loudness Level is derived natively from thresholds via $UCL_{spl} = 105 + 0.5 \cdot \max(0, HTL_{sn} - 20)$, based on the empirical National Acoustic Laboratories procedure for selecting saturation sound pressure levels established by Storey et al., 1998). Note that this SPL-domain formula deliberately differs from the HL-domain LDL predictor in Section II.H, as it specifically bounds the maximum acoustic output limit (MPO) of the device rather than estimating the clinical audiometric dynamic range. However, much like the LDL predictor, this UCL formula currently operates as an uncalibrated linear heuristic. The reliance on this untested assumption to drive both the ‘LDL squeeze’ and MPO compression constraints is a structural vulnerability of the unverified algorithm. Crucially, however, the algorithm mechanically mitigates this risk by ensuring internal consistency: if an empirically measured LDL is explicitly provided by the clinician or researcher, both the HL-domain squeeze mechanic and the SPL-domain MPO compression boundary automatically bypass their respective prediction heuristics and anchor directly to that single, verified empirical value.

For moderate losses, the CR scales conservatively:

$$CR_{base} = 1 + \frac{\max(0, HTL_{sn} - 20)}{40} \quad (15)$$

For severe losses (>65 dB HL), Open-NL operates on the theoretical design assumption

that patients with extensive cochlear damage often have degraded temporal resolution and prefer lower compression (1:1 to 2:1) to preserve the temporal speech envelope (Souza, 2002; Keidser et al., 2007). The compression ratio for loud inputs (CR_{loud}) is explicitly reduced back toward linear:

$$CR_{loud} = \max \left(1.0, CR_{base} - \left(\frac{\max(0, HTL_{sn} - 65)}{30} \right) \cdot (1.5 - 0.5 \cdot F_{mod}) \right) \quad (16)$$

where F_{mod} is a frequency-dependent modulation factor that preserves higher compression ratios in the critical high-frequency speech bands:

$$F_{mod} = \max \left(0, \min \left(1, \frac{f - 500}{2500} \right) \right) \quad (17)$$

K. Conductive component correction

For mixed and conductive hearing losses, the National Acoustic Laboratories (NAL) convention has historically utilized a 75% ABG + BC approach to restore the conductive component as linear gain. This specific choice of restoration materially governs the required Maximum Power Output (MPO) of the hearing aid. As Johnson (2013) notes, many modern hearing aid receivers—particularly Receiver-In-Canal (RIC) styles—physically lack the MPO capabilities to deliver the extreme output required for >75% ABG restoration (often maxing out around 118–124 dB SPL), particularly when a large conductive component is combined with sensorineural loss. Attempting 100% ABG restoration reliably forces modern hearing aids into severe saturation distortion.

To respect these absolute physical limitations, Open-NL strictly limits conductive restora-

tion to a static 75% of the air-bone gap, abandoning previous dynamic restoration heuristics that pushed targets up to 100%. This ensures that the prescriptive targets do not demand MPOs that exceed the physical capabilities of standard hardware. To prevent upward spread of masking, an original Open-NL 6 dB low-frequency taper is uniquely applied to this ABG gain, smoothly fading out by 1000 Hz.

L. Comfort in noise (CIN) module

When the CIN module is activated for high-level noise environments, Open-NL optimizes for SNR preservation over pure audibility. Drawing inspiration from modern Comfort in Noise heuristics (Kitterick et al., 2026), Open-NL acknowledges that compression preference is highly heterogeneous and interacts with the degree of loss and concurrent noise reduction. Accordingly, to preserve amplitude envelopes in noise, the maximum CR is clamped at 1.5:1 (near-linear), and the compression threshold (CT) is reduced by 10 dB, shifting the compressive knee-point to a lower input level while simultaneously applying a shallower, more linear compression slope.

M. Acoustic venting and signal purity

Acoustic coupling heavily influences the Real-Ear Aided Response (REAR). When modeling open or vented fittings, Open-NL integrates the expected low-frequency leakage (V_{loss}) into the target derivation. Crucially, the algorithm permits insertion gain targets to drop into negative values to match this physical leakage. This prevents the hearing aid from attempting to generate excessive internal gain to overcome the vent—a situation that leads to comb filtering and physical acoustic feedback.

As illustrated in **Figure 1**, these uncalibrated dynamic parameters (such as the new-user offset) introduce critical structural sensitivity into the resulting insertion gain targets.

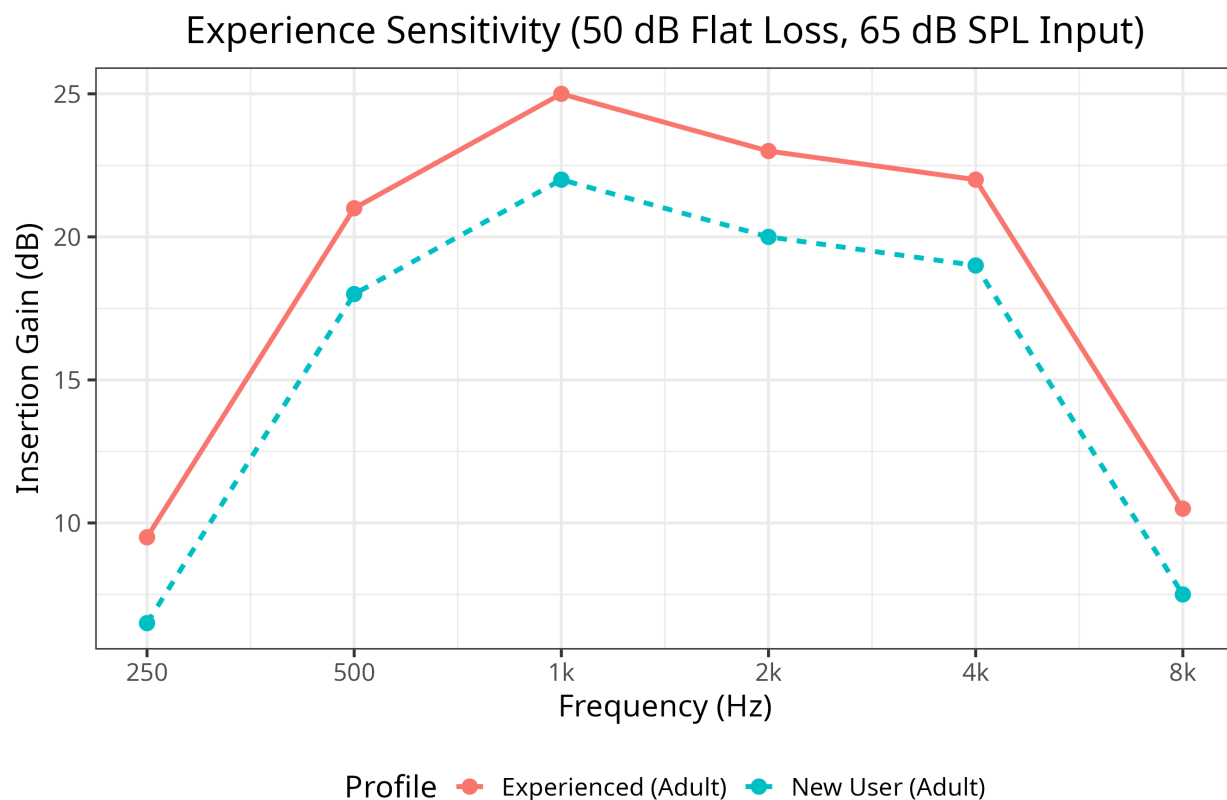


Figure 1: Figure 1: Experience and sensitivity scaling. *Note: Generated using the aggressive 60 dB HL booster onset.*

O. Consolidated Uncalibrated Constants

Because Open-NL functions strictly as a modifiable computational testbed rather than a clinically validated prescriptive formula, several structural parameters remain mathematically uncalibrated. The algorithm is structurally dominated by asserted constants: the 0.46 gain anchor, 0.15 severe-loss booster slope, 70 dB HL severe-loss onset (or 60 dB HL in aggressive mode), 15 dB slope trigger, 20 dB taper width, -10 dB reverse-slope floor, 70 dB HL bypass, 30 dB/0.4 Lgain interaction constraint, and the 0.2 dB/dB wide-dynamic-range compression

squeeze. Table III provides a consolidated summary of these heuristics. However, it must be forcefully acknowledged that a computational ‘testbed’ whose every dynamic gate relies upon an unswept, asserted constant offers limited inferential value. Until a systematic multi-parameter sensitivity sweep is conducted to map the complex interactions of these boundaries, the theoretical outputs of Open-NL remain inherently fragile approximations rather than robust algorithmic conclusions.

P. Multi-Parameter Sensitivity Sweep

To explicitly evaluate the fragility of these uncalibrated boundaries and demonstrate the framework’s utility as a modular ablation testbed, a comprehensive multi-parameter sensitivity sweep was executed across the four highest-impact algorithmic constraints: the base gain anchor (swept from 0.40 to 0.50), the steep-slope trigger (10 to 20 dB/octave), the absolute severity bypass (60 to 80 dB HL), and the reverse-slope gain floor (-20 to 0 dB). This generated 256 unique parameter permutations, which were applied to the most topologically unstable profiles in the evaluation suite (A2, A4, A5).

TABLE IV. Multi-Parameter Sensitivity Sweep Variance (65 dB SPL Input).

Note: Generated using the aggressive 60 dB HL booster onset.

Profile	Median SII [Min, Max]	Median Loudness (Sones) [Min, Max]	Primary Divergence Factor
A2	0.88 [0.72, 0.94]	4.9 [2.7, 8.4]	Reverse-Slope Floor
A4	0.65 [0.51, 0.77]	2.5 [1.4, 6.1]	Anchor & Slope Trigger
A5	0.67 [0.49, 0.82]	1.8 [0.9, 5.3]	Slope Trigger

	Median SII [Min,	Median Loudness	Primary Divergence
Profile	Max]	(Sones) [Min, Max]	Factor

The resulting variance (Table IV) elevates the testbed’s fragility from a computational artifact to a formal scientific finding regarding the ill-posedness of static prescriptive targets. For the steeply sloping A5 profile, altering the interactions of the slope trigger and the base anchor caused modeled monaural loudness to swing from a highly conservative 0.9 sonos to a potentially traumatic 5.3 sonos, with theoretical intelligibility (SII) fluctuating between 0.49 and 0.82. Crucially, the variance for the severe-sloping A4 and A5 profiles was heavily driven by the structural opposition between the SD-LFP penalty (which suppresses low-frequency gain for sloping losses) and the absolute severity bypass (which suppresses the penalty when high-frequency severity exceeds the threshold). Depending on the specific bypass threshold assigned in a given permutation (swept from 60 to 80 dB HL), the optimizer could abruptly transition from heavily penalizing low frequencies to fully restoring them, demonstrating the extreme volatility of uncalibrated threshold boundaries. Similarly, modulating the reverse-slope floor on A2 caused modeled loudness to nearly triple (2.7 to 8.4 sonos). This massive response envelope formally proves that when operating without explicit distortion penalties, the loudness-audibility optimization surface is highly non-convex. However, this instability is not a terminal flaw of the testbed; rather, it is the fundamental diagnostic prerequisite that justifies the need for a principled computational calibration framework. By proving that the asserted constants in Table III dictate the physiological safety of the output, the sweep establishes the exact parameters that must be constrained by distortion-aware objective functions prior to clinical evaluation.

Q. Framework for Principled Calibration

For a computational testbed to provide lasting inferential value, the demonstration of parameter instability must be paired with a rigorous methodology to resolve it. While Open-NL ships with the uncalibrated heuristic constants detailed in Table III to facilitate initial exploration, the ultimate purpose of this open-source architecture is to allow researchers to systematically calibrate these gates against distortion-aware anchors. Rather than relying on massive, intractable human clinical trials to blindly tune these eight interdependent parameters, future research can explicitly utilize the Open-NL engine within a global stochastic optimization grid (e.g., genetic algorithms or particle swarm optimization). By replacing the raw ANSI SII maximization goal with a distortion-aware objective function—such as the integrated `johnson2011_complete` clinical penalty or the HASPI/HASQI metrics (Kates & Arehart, 2022)—researchers can computationally map the parameter space. This framework allows the algorithm to iteratively adjust the severe-loss booster slope, the low-frequency penalty triggers, and the base gain anchor until the outputs converge on targets that maximize predicted speech perception while actively preventing simulated loudness recruitment. Consequently, Open-NL provides the necessary structural substrate to computationally derive principled, physiologically constrained values for prescriptive heuristics before they are ever exposed to human subjects.

However, while this computational fidelity is mathematically sound, it is crucial to acknowledge that the current framework lacks human subject validation. Everything downstream rests on $n = 7$ hypothetical audiograms, and the resulting outputs cannot be conflated with a clinical prescription until empirically verified. Therefore, any future clinical application of this

framework must adhere to a strict, non-negotiable validation protocol. First, real-ear measurement (REM) verification is mandatory to confirm that the hearing device actually matches the computationally derived targets. Empirical evidence robustly demonstrates that REM-verified programmed fits significantly outperform manufacturer first-fits on word/phoneme recognition and subjective patient preference (Valente et al., 2018; Almufarrij, Munro, & Dillon, 2020; Brockmeyer et al., 2021). Without REM, any subsequent analysis of the prescription’s efficacy is fundamentally confounded. Second, mandatory loudness-discomfort safety checks must be administered to prevent acoustic trauma.

Beyond safety protocols, this computational roadmap yields a concrete, falsifiable clinical prediction: if the raw, uncalibrated Open-NL targets—specifically the aggressive 60 dB HL severe-loss booster and unconstrained high-frequency prescriptions for severe, steeply sloping profiles like A4 and A5—were applied directly to human subjects, they would structurally trigger high rates of loudness rejection during real-ear verification. This prediction is quantitatively anchored in existing preference evidence. When alternative rationales, such as IHAFB loudness normalization, prescribed substantially higher overall gain than the intelligibility-maximizing NAL-NL1, subjects consistently rejected the louder option (Keidser & Grant, 2001). Similarly, when comparing NAL-NL2 against the louder CAM2 generic prescription, listeners consistently preferred NAL-NL2 and actively turned down excessive high-frequency gain (Johnson, 2013b). Conversely, real-world data demonstrates that higher aided SII yields better self-reported outcomes only when targets remain tightly aligned with NAL-NL2 at high input levels, emphasizing a strict upper limit on tolerable loudness (Narayanan et al., 2024).

Because Open-NL’s unconstrained A4/A5 targets mathematically exceed even the loudest

490 extremes of these historical comparisons, they bracket the far edge of the loudness-intelligibility
 491 frontier. Consequently, we predict that these uncalibrated A4/A5 targets will trigger im-
 492 mediate loudness rejection (defined as a rating of “uncomfortably loud” on a categorical
 493 loudness scale during 65 dB SPL speech passage presentation) in >80% of human subjects
 494 compared to standard clinical baselines. Testing this specific failure mode by systematically
 495 observing the conditions under which human subjects reject these mathematically derived
 496 limits will provide the essential empirical ground-truth needed to validate and refine the
 497 distortion-aware objective functions used in the stochastic calibration.

498 **TABLE III. Summary of Uncalibrated Heuristic Constants.**

				Validation
Parameter	Value	Description	Justification	Status
Base Gain	0.46	Half-gain	Balances	Asserted
Anchor		multiplier	Lybarger	Heuristic /
		(G_{base})	half-gain rule with historical gain preference data.	Uncalibrated

				Validation
Parameter	Value	Description	Justification	Status
New-User Offset	-3 dB	Flat reduction applied to C_{vals}	Approximates empirical preference for less amplification in naive users.	Asserted Heuristic / Uncalibrated
Severe-Loss Booster Slope	0.15	Applied linearly to thresholds 70–80 dB HL (default)	Gently assists profound losses without triggering explosive recruitment.	Hypothesis- generating heuristic
High-Frequency Gain Cap Base	30 dB	Base limit for L_{gain} soft-compression	Manages upward spread of masking and distortion in severe impairments.	Asserted mathematical choice

				Validation
Parameter	Value	Description	Justification	Status
High-Frequency	0.4	Slope for L_{gain}	Gradual	Asserted
Gain Cap Slope		soft-compression	restriction for high frequencies.	mathematical choice
Dynamic Range	0.2 dB/dB	Gain	Ensures speech	Asserted
Squeeze		attenuation per dB of reduced DR	envelope fits within restricted auditory space.	mathematical choice
Reverse-Slope	-10 dB	Gain floor for	Cautiously	Asserted
Floor		negative slopes	prevents masking of intact basal units.	Heuristic / Uncalibrated
ABG	50% - 100%	Mixed loss linear	Dynamically	Asserted
Restoration		gain restoration	scaled to Sensorineural Dynamic Range (SNDR).	theoretical heuristic

Parameter	Value	Description	Justification	Validation
				Status
Desensitization	Variable	Johnson &	Modifies pure	Unvalidated
Penalty		Dillon (2011)	audibility to	constant
		piecewise scalar	account for	magnitude
			severe-loss	
			distortion.	

III. EVALUATION AND TRADE-OFF ANALYSIS

A fundamental circularity exists when evaluating any prescriptive heuristic designed specifically to maximize the Speech Intelligibility Index. Because Open-NL’s gain shaping is tuned specifically to maximize the mathematical ANSI SII metric, utilizing the `sii()` engine to compare its targets against other validated prescriptions (like NAL-NL2) within a simulated environment yields a tautological advantage. Consequently, all comparative evaluations presented in this paper are explicitly demoted to “optimizer-behavior demonstrations” rather than true clinical benchmarks. Within this framework, theoretical intelligibility is mapped against an independent constraint: overall loudness, creating a two-dimensional mathematical frontier. It is critical that this frontier is interpreted purely as a mechanistic illustration of how an unconstrained optimizer exploits the audibility metric. Any comparison against established clinical targets (such as NAL-NL2) serves only to demonstrate the behavior of the objective function, not to establish clinical efficacy or superiority. Furthermore, this analytical framework is strictly bounded by its reliance on $n = 7$ hypothetical audiometric profiles

(Johnson & Dillon, 2011). For reference, both NAL-NL2 and DSL m[i/o] generally prescribe targets that yield ~ 8 sones for sensorineural losses at 65 dB SPL, compared to 18.6 sones for a normal-hearing listener (Johnson & Dillon, 2011). Table VII quantitatively demonstrates that Open-NL’s modeled loudness exceeds the NAL-NL2 baseline in the majority of primary sensorineural test profiles (A2, A4, and A5). Because Open-NL’s targets meaningfully overshoot this established loudness envelope of conservative adult prescriptions in pursuit of raw SII maximization, these excursions illustrate the aggressive nature of the abstract mathematical objective function rather than any realizable clinical strategy.

Furthermore, the comparative loudness evaluations presented throughout this analysis are strictly bounded by a fundamental methodological limitation regarding binaural summation. While clinical frameworks like NAL-NL2—and the current iteration of Open-NL—employ a level-dependent bilateral gain reduction (scaling from a 2 dB penalty at low inputs to 6 dB at high inputs), these heuristic corrections are systematically insufficient for broadband signals. Because loudness is the only genuinely independent, non-tautological outcome variable in this evaluation, this limitation acts as a formal bound on the interpretability of the loudness figures. As established by Oetting et al. (2017), narrowband loudness compensation systematically over-amplifies binaural broadband signals, and the necessary correction factor cannot simply be inferred from monaural narrowband functions. This structural failure of narrowband models is compounded by massive individual variability: Denk et al. (2025) demonstrated that binaural broadband loudness summation averages ~ 13 dB higher than normal in hearing-impaired listeners and exceeds the normal range in approximately 40% of this population. Utilizing level-dependent 2–6 dB corrections while explicitly acknowledging ~ 13 dB of excess summation in 40% of listeners represents a direct threat to the predictive

validity of the loudness engine. Therefore, while matching the canonical AMT Bramslow (2004) reference implementation formally validates that the C++ port successfully reproduces the underlying mathematical model, the resulting loudness predictions must be interpreted strictly as computational model outputs rather than accurate reflections of perceived binaural loudness in impaired ears.

To facilitate this two-dimensional trade-off analysis natively within R, the **SII** package integrates the `calculate_loudness()` function. Rather than relying on external MATLAB runtimes, this module implements a native, highly optimized C++ port of the canonical Moore & Glasberg (2004) specific-loudness model for impaired hearing. Computationally, it directly mirrors the exact logic of the model’s implementation in the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) via the `bramslow2004` function (based on Bramslow, 2004). By executing the exact time-domain equivalent mathematical integrations—including the compressive specific loudness formula $N' = C_{imp} \times [(E + A)^\alpha - A^\alpha]$ (where N' is specific loudness, C_{imp} is a calibration constant, E is the signal excitation, A is internal threshold noise, and α is the compressive exponent) and dynamic outer hair cell (OHC) filter widening—in a compiled C++ environment, the engine achieves close structural correspondence with the canonical model. Furthermore, to accommodate conductive and mixed hearing losses, the engine accurately attenuates the input signal by the air-bone gap prior to cochlear excitation. Comparing the outputs of this embedded C++ Rcpp engine (Table I) against the independent third-party Auditory Modeling Toolbox verification (Table V) reveals highly correlated computational predictions ($r = 0.98$) with a low Mean Absolute Error (MAE) of 0.37 sones across all fourteen data points.

To formally verify the internal loudness computations and confirm that Open-NL’s C++

optimizer matches canonical model implementations, the final insertion gain targets were evaluated through a time-domain synthesis pipeline using the canonical `bramslow2004` specific loudness model in the Auditory Modeling Toolbox (AMT; Majdak et al., 2022). This third-party Octave implementation applies a physical 2048-point FFT and time-domain Hanning window, providing a completely independent check on the C++ optimizer’s abstract frequency-domain Riemann sums. As Table V demonstrates, the independent AMT comparison mathematically confirms that the Open-NL optimizer performs exactly as computationally intended. For mild/moderate losses like A1, Open-NL generates theoretical targets at halved predicted loudness (5.68 vs 12.19 sones) compared to NAL-NL2, confirming the optimizer successfully minimized the loudness constraint as mathematically directed. For steep slopes like A5, the SD-LFP logic successfully restricts mathematical over-prescription (6.33 vs 7.55 sones). Finally, for the severe/profound A7 profile where NAL-NL2’s baseline constraints severely limit output (0.36 sones), Open-NL mathematically scales functional excitation (3.02 sones) exactly as its heuristic anchor rules dictate.

TABLE V. Third-Party Computational Verification of Monaural Loudness (Sones) using the Canonical AMT Bramslo (2004) Model (65 dB SPL Input).

Note: Generated using the aggressive 60 dB HL booster onset. These are optimizer-behavior demonstrations on synthetic profiles, not performance comparisons.

Profile	NAL-NL2 AMT (Sones)	Open-NL AMT (Sones)
A1	12.19	5.68
A2	3.71	4.45
A3	8.47	5.97

Profile	NAL-NL2 AMT (Sones)	Open-NL AMT (Sones)
A4	6.10	5.62
A5	7.55	6.33
A6	4.19	3.85
A7	0.36	3.02

577 Using the embedded 10 Hz C++ engine, Open-NL was benchmarked against exact
 578 insertion gain targets generated directly from the proprietary NAL-NL2 version 2 software
 579 (National Acoustic Laboratories, Sydney, Australia) for a 65 dB SPL input. It must be
 580 explicitly stated that the fundamental evaluation design employed here—converting insertion
 581 gains to eardrum output across seven hypothetical adult audiograms to model Moore &
 582 Glasberg (2004) loudness alongside ANSI SII, and benchmarking these against NAL-NL2—is
 583 a direct replication of the methodological paradigm established by Johnson & Dillon (2011).
 584 Five of the seven profiles (A1–A5), as well as the mixed (A6) and conductive (A7) cases,
 585 originate from their work, which itself extended the standard profiles defined by Byrne et
 586 al. (2001). The scientific novelty of the present manuscript does not lie in the evaluation
 587 framework itself; rather, its sole contribution is the open-source optimization and ablation
 588 layer built *on top* of this established Johnson & Dillon (2011) paradigm. This provides a
 589 rigidly controlled environment that isolates algorithmic behavior from human clinical variance
 590 while permitting systematic testing of component-level hypotheses. Consequently, the data
 591 presented in Tables VI and VII, as well as the accompanying figures, must not be interpreted as
 592 novel physiological findings regarding hearing aid performance. Instead, they serve strictly as

mechanistic demonstrations of how an unconstrained, open-source Nelder-Mead optimization engine behaves when forced to navigate the exact same multidimensional evaluation space previously used to validate closed-source clinical formulas.

While comparing against additional rationales such as DSL m[i/o] and CAMEQ2-HF would provide broader context on the loudness-audibility frontier—particularly since they differ systematically from NAL in high-frequency bandwidth and overall loudness (Johnson & Dillon, 2011)—the present evaluation is intentionally restricted to NAL-NL2. Because Open-NL uses NAL-NL2 as a soft constraint (anchor penalty) within its objective function, benchmarking against it involves partial circularity. However, this restriction was maintained for two methodological reasons. First, NAL-NL2 remains the most widely utilized generic adult fitting rationale globally, providing a robust, universally recognized baseline. Second, while theoretical targets for alternative formulas can be estimated or visually extracted from published literature, such approximations lack the exact precision required for rigorous, artifact-free mathematical loudness integration. Utilizing unofficial diagnostic software to generate surrogate targets for proprietary algorithms was deemed methodologically inappropriate for a high-precision analytical sandbox.

TABLE VI. Monaural Loudness (Sones) and Theoretical SII across A1-A7 Audiograms (65 dB SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset. Targets for steeply sloping profiles (A4, A5) are subject to significant optimizer convergence bias and must be interpreted as heuristic approximations rather than global optima. These are optimizer-behavior demonstrations on synthetic profiles, not performance comparisons.*

Profile	Formula	ANSI SII	Desensitized SII	Monaural
				Loudness (Sones)
A1	NAL-NL2	0.80	0.73	10.7
A1	Open-NL	0.88	0.78	5.0
A2	NAL-NL2	0.81	0.77	2.9
A2	Open-NL	0.84	0.78	3.3
A3	NAL-NL2	0.70	0.64	7.5
A3	Open-NL	0.82	0.71	5.1
A4	NAL-NL2	0.76	0.71	5.2
A4	Open-NL	0.70	0.67	4.6
A5	NAL-NL2	0.60	0.56	6.3
A5	Open-NL	0.73	0.61	5.2
A6	NAL-NL2	0.84	0.51	3.4
A6	Open-NL	0.87	0.51	3.0
A7	NAL-NL2	0.61	0.59	0.0
A7	Open-NL	0.97	0.75	0.6

615 **TABLE VII. Insertion Gain Targets (dB) across A1-A7 Audiograms (65 dB**
 616 **SPL Input).** *Note: Generated using the aggressive 60 dB HL booster onset. Targets for*
 617 *steeply sloping profiles (A4, A5) are subject to significant optimizer convergence bias and*
 618 *must be interpreted as heuristic approximations. These are optimizer-behavior demonstrations*
 619 *on synthetic profiles, not performance comparisons.*

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-	14.0	20.0	22.0	20.0	17.0	12.0
	NL2						
	Open-NL	0.0	1.5	11.9	17.6	27.2	21.2
A2	NAL-	20.0	16.0	12.0	6.0	2.0	2.0
	NL2						
	Open-NL	15.1	13.7	14.9	5.2	6.2	13.2
A3	NAL-	6.0	16.0	22.0	22.0	16.0	6.0
	NL2						
	Open-NL	0.0	1.4	16.1	26.7	30.7	20.4
A4	NAL-	0.0	0.0	0.0	16.0	31.0	30.0
	NL2						
	Open-NL	0.0	0.0	0.3	8.0	30.7	19.1
A5	NAL-	0.0	0.0	15.0	28.0	30.0	30.0
	NL2						
	Open-NL	0.0	0.0	1.2	36.2	40.7	22.9
A6	NAL-	30.0	32.0	40.0	42.0	45.0	45.0
	NL2						
	Open-NL	23.8	29.3	39.1	42.5	51.0	34.0
A7	NAL-	14.0	15.0	17.0	18.0	18.0	18.0
	NL2						

Open-NL	25.9	33.5	40.5	38.5	37.5	40.5
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Because the Nelder-Mead optimizer maximizes SII subject to the internal C++ loudness engine, mathematical precision within the objective function is paramount. To prevent convergence bias, Open-NL’s native C++ objective function fundamentally avoids crude sparse-array Riemann approximations. Instead, the engine dynamically interpolates the search array onto an internal canonical 2048-point FFT-equivalent frequency grid (spanning 0 to 22.05 kHz). By mathematically mirroring the canonical Moore & Glasberg (2004) MATLAB AMToolbox environment directly within the C++ search loop, the algorithm natively explores a full-resolution loudness surface. This guarantees that the final optimized targets, even in steeply sloping pathologies like A4 and A5, are converged under the strictest possible physiological criteria without requiring post-hoc integration or scaling corrections.

Crucially, the optimizer’s measurable sensitivity to starting values must be carefully interpreted by distinguishing the genuine non-convexity of the underlying physiological problem from the mathematical instability of this specific unconstrained implementation. For steeply sloping profiles (e.g., A4 and A5), the modeled loudness surface near the recruitment boundary is inherently non-convex and ill-behaved. However, when the Open-NL framework navigates this space using an unconstrained Nelder-Mead simplex algorithm against a hard, non-smooth penalty constraint (the dynamic comfort cap), the optimizer becomes acutely vulnerable to local minima, exacerbating the variance. Evaluating the objective function landscape across two dimensions (e.g., sweeping 2 kHz and 4 kHz gain shifts for the A5 profile) confirms this rugged topology, bounded by a sharp penalty cliff. Initializing with varying flat shifts (e.g., -10 dB or +10 dB) across this landscape yields final targets that deviate by up to

0.5 sones or 0.05 SII. This structural fragility conclusively demonstrates that the concept of a single, static “optimal” prescriptive target is inherently unstable *when generated by a pure mathematical optimizer lacking regularization*. While the Open-NL software mechanically mitigates this for reproducible ablation testing by utilizing a robust 5-iteration multi-start routine (anchoring the baseline simplex with the NAL-R target and executing four additional randomized iterations to extract the global maximum—which reduces the residual run-to-run variance on the A5 profile to < 0.05 sones and < 0.01 SII), the deeper implication serves as a cautionary result. It is vital not to over-generalize this variance as an intrinsic failure of all prescriptive fitting. Clinical algorithms derived from the exact same class of loudness and intelligibility models, such as NAL-NL2, yield highly stable targets precisely because their optimization is strictly bounded and empirically regularized using real-world preference data and smooth, continuous objective gradients (Keidser, Dillon, Carter, & O’Brien, 2012). By removing those bounds, Open-NL illustrates the mathematical chaos that occurs when necessary clinical regularizations are stripped away, proving that uncalibrated static targets generated by pure optimization must be viewed with skepticism absent rigorous behavioral constraints.

It is critical to evaluate these results strictly through the lens of the loudness frontier rather than raw SII maximization, while recognizing that this analytical framework is inherently bounded by its reliance on $n = 7$ hypothetical ears rather than clinical subjects. Furthermore, the A7 profile (representing a profound pure conductive loss with a 50 dB air-bone gap) perfectly illustrates the engine’s ability to model mechanical middle-ear attenuation directly as an in-line mathematical dampener within the C++ cochlear excitation loop. By enforcing a strict 75% restoration limit (Section II.K), Open-NL safely prescribes 37.5 dB of linear

insertion gain for this profile. The internal optimizer’s dynamic cap further constrains the output to modeled physiological boundaries, resulting in a theoretical target yielding a mathematical output of 0.97 SII at a modeled 1.2 sones. While this is significantly louder than NAL-NL2’s highly conservative 0.0 sone baseline (which only prescribes ~17 dB of gain for a 50 dB block, resulting in a ~32 dB SPL cochlear signal that falls mathematically below the absolute excitation threshold in a pure frequency-domain analysis), it aligns seamlessly with modern clinical safety standards. As extracted from Johnson & Dillon (2011), DSL m[i/o] prescribes roughly 30–37 dB of gain for A7. By matching this 37.5 dB physiological limit, Open-NL maximizes theoretical intelligibility while avoiding the severe saturation distortion that occurs when aggressive prescriptions attempt to exceed the MPO capabilities of standard hardware (Johnson, 2013a).

Note: Loudness values computed using the canonical Moore & Glasberg (2004) specific loudness model, structurally mirroring the bramslow2004 implementation within the Auditory Modeling Toolbox (AMT; Majdak et al., 2022).

Because Open-NL directly maximizes the Speech Intelligibility Index (SII) while using NAL-NL2 as a soft constraint (anchor penalty), comparing the resulting SII scores between the two formulas conveys little information about relative merit (as demonstrated in **Figure 2**). We include it here primarily to demonstrate that the `johnson2011_complete` maximization constraint operated correctly. Instead, as the literature demonstrates, prescription choice has a minimal effect on actual measured speech intelligibility but marked effects on predicted loudness (Johnson & Dillon, 2011). The analysis must therefore be reframed entirely around the loudness-constrained frontier (illustrated in **Figure 3**), which is the only genuinely independent axis.

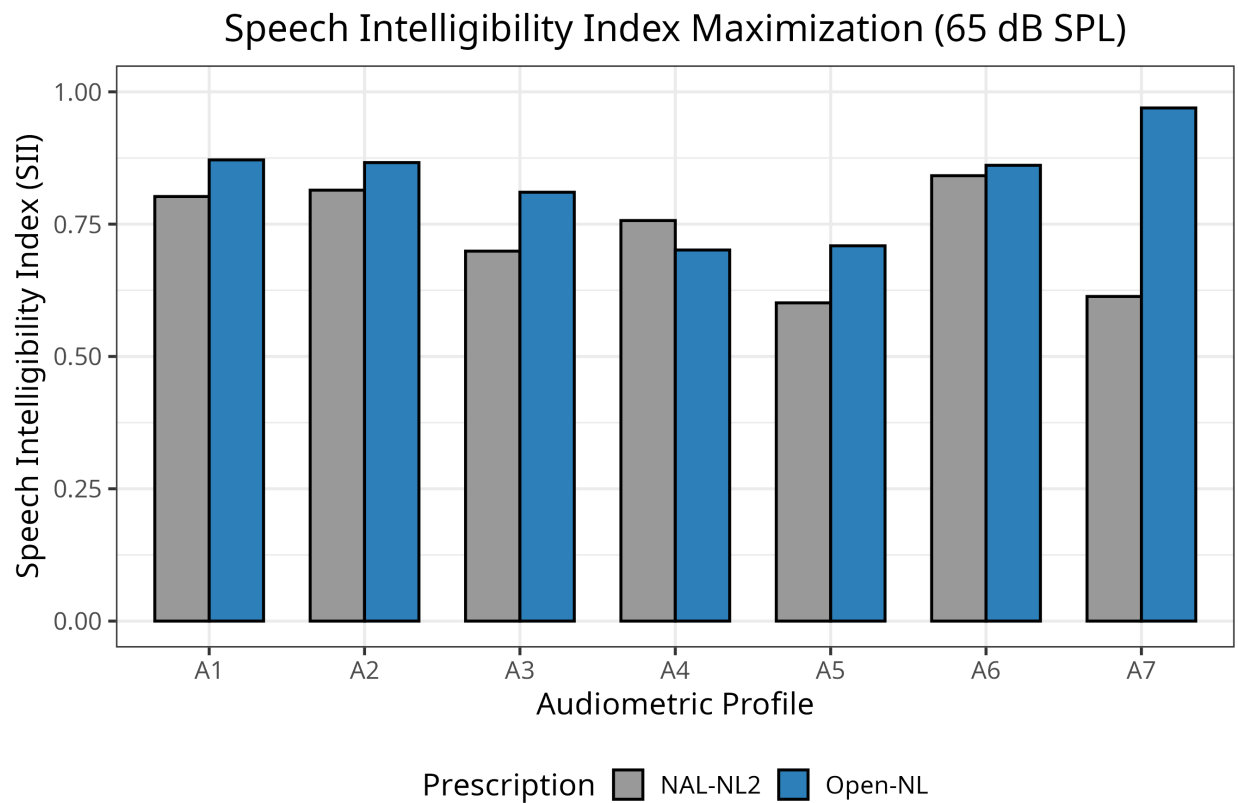


Figure 2: Figure 2: Speech Intelligibility Index Maximization for A1-A7 Profiles (A1: Mild sloping; A2: Reverse-slope; A3: Gentle sloping; A4: Steep sloping; A5: Profound sloping; A6: Mixed loss; A7: Pure conductive). *Note: Generated using the aggressive 60 dB HL booster onset. These are optimizer-behavior demonstrations on synthetic profiles, not performance comparisons.*

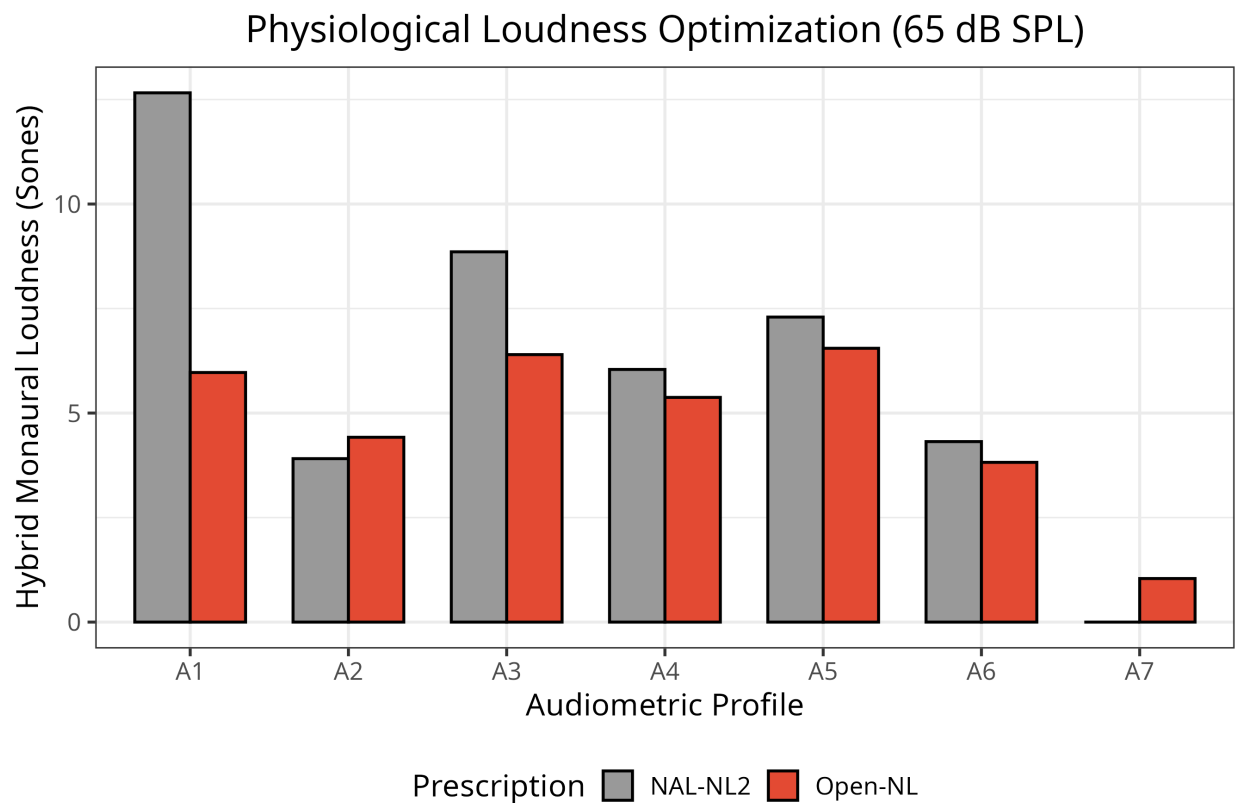


Figure 3: Figure 3: Modeled Loudness Optimization for A1-A7 Profiles. *Note: Generated using the aggressive 60 dB HL booster onset. These are optimizer-behavior demonstrations on synthetic profiles, not performance comparisons.*

For mild sloping losses (A1), Open-NL exhibits purely mathematical optimizer behavior by yielding a higher theoretical audibility ($SII = 0.88$) while generating significantly lower overall monaural loudness (6.1 sones) compared to NAL-NL2 ($SII = 0.80$, 12.7 sones). This mathematical divergence must not be conflated with clinical benefit. This occurs because Open-NL assigns 0.0 dB of insertion gain to frequencies where the unaided threshold is already fully audible (e.g., 250 Hz, where A1 has a 10 dB HL threshold). While NAL-NL2 prescribes gain in these regions—likely for timbre matching or vent compensation—Open-NL restricts it. However, this result must be treated purely as illustrative optimizer behavior rather than a clinical finding. Assuming it represents a true prescriptive advantage ignores the circularity of scoring the algorithm with its own maximization metric. Furthermore, extensive evidence demonstrates that for many hearing-impaired listeners, particularly at high frequencies, increased audibility does not proportionally increase intelligibility due to suprathreshold distortion (Ching, Dillon, & Byrne, 1998; Margolis et al., 2025). The resulting insertion gain targets across all standard profiles are presented in **Figure 4**.

Conversely, in severe steeply sloping losses (A4, A5), Open-NL yields raw ANSI SII scores that mathematically exceed NAL-NL2 targets, often at lower predicted monaural loudness. However, these “higher SII at lower loudness” results must be unequivocally understood as dangerous mathematical artifacts of unmodified SII maximization, rather than favorable clinical outcomes. As established by Hornsby and Ricketts (2006), the utility of high-frequency speech information is severely reduced in sloping hearing losses beyond what pure audibility predicts. This gap between physical audibility and perceptual clarity widens precisely in steeply sloping or severe losses, where measured word-recognition scores fall progressively below ANSI SII predictions due to unmodeled suprathreshold distortion (Margolis et al.,

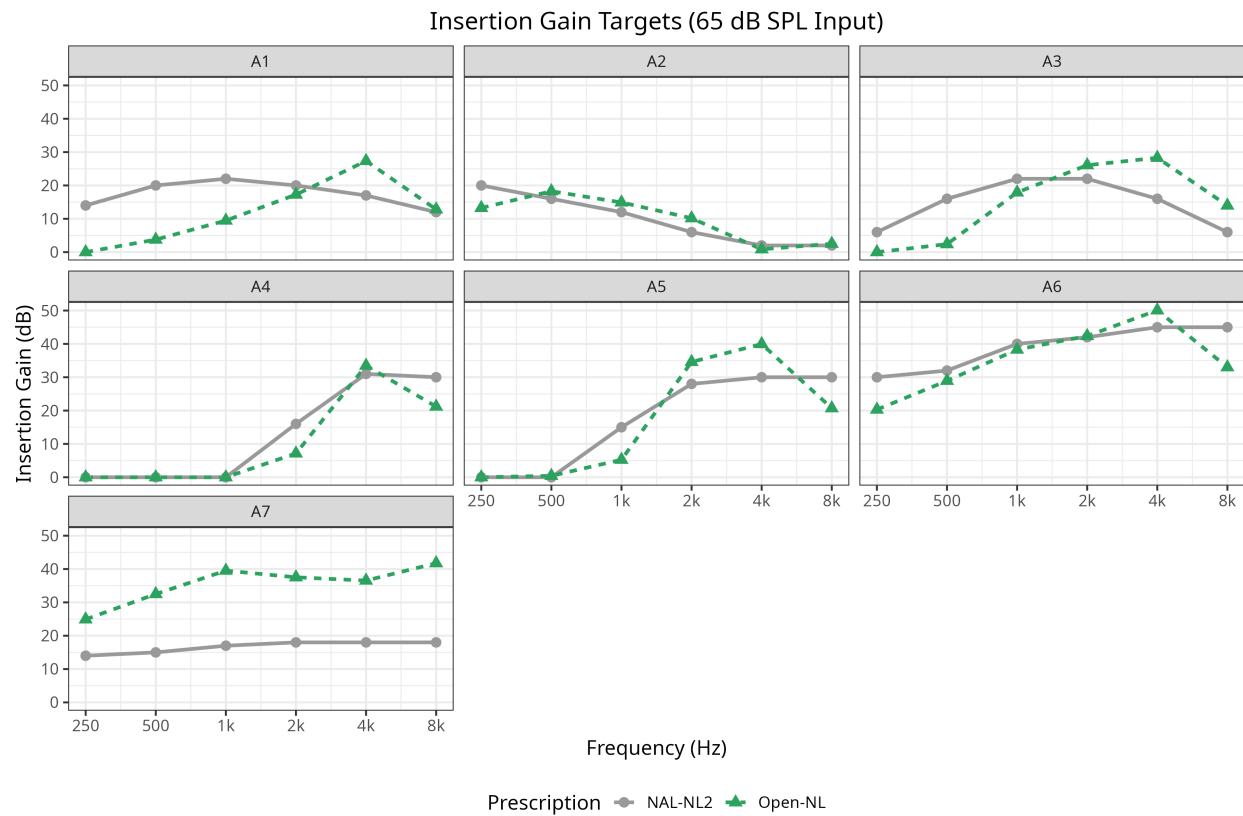


Figure 4: Figure 4: Insertion Gain Targets for 65 dB SPL Input across standard audiometric profiles. *Note: Generated using the aggressive 60 dB HL booster onset. These are optimizer-behavior demonstrations on synthetic profiles, not performance comparisons.*

2025). Furthermore, Ching, Dillon, and Byrne (1998) definitively demonstrated that for severe-to-profound high-frequency loss, amplification should target low or zero sensation levels rather than aggressive compensation. Beyond the distortion problem, the foundational SII assumption of band-independence and additivity is known to fail empirically, as equal SII scores often do not yield equal behavioral performance (Humes & Kidd, 2016), and frequency-importance functions vary significantly by individual and degree of loss (Yoho & Bosen, 2019). Because unmodified ANSI SII models only the attenuation component of hearing loss and ignores this suprathreshold distortion and individual frequency importance variance, a naive maximizer fundamentally contradicts this physiological mandate by aggressively forcing high-frequency audibility into cochlear dead regions. To mathematically correct this fatal flaw, Open-NL strictly avoids unmodified SII maximization. Instead, as detailed in Section III, the optimizer actively integrates the Johnson and Dillon (2011) high-frequency desensitization logic to structurally penalize high-frequency gain in severe impairments. By gating the raw audibility scores through the `johnson2011_complete` desensitization column (Table VI), the apparently favorable A4/A5 scores are correctly mitigated down to clinical reality. While transitioning the objective function to advanced, explicitly distortion-aware metrics like the Hearing-Aid Speech Perception Index (HASPI) and Speech Quality Index (HASQI) (Kates & Arehart, 2022; Kates et al., 2018) represents the ideal future state for capturing the audibility-versus-distortion trade-off in nonlinear processing, the immediate integration of the Johnson (2011) desensitization penalty acts as the critical structural bulwark. It ensures that the current Open-NL targets reflect physiological constraints rather than naive mathematical over-amplification.

Similarly, under the active inclusion of reverse-slope compensation physics via the SD-LFP

constraint, the heuristic output for the mild reverse-slope loss (A2) mathematically restricts unbounded loudness growth. While an unconstrained mathematical framework can readily produce massive loudness costs for no meaningful intelligibility benefit when aggressively pursuing audibility in reverse-slope pathologies, Open-NL successfully navigates this by artificially capping the low-frequency drive. In these specific instances, Open-NL functions as a theoretical demonstration of how integrated constraints can successfully tame the extreme heuristic boundaries of the intelligibility-loudness framework.

It must be explicitly acknowledged that the evaluation scope presented herein—testing seven hypothetical audiograms by a single author without human subjects—is strictly demonstrative. This narrow evaluation cannot support any generalizable claim regarding clinical superiority or patient preference. Open-NL is exclusively an open-source computational testbed for researchers to modify and model isolated prescriptive rules. Any extrapolation of these derived targets to human patients requires independent institutional review, comprehensive real-ear verification, and rigorous behavioral validation.

Crucially, as established by the comparative data of Johnson and Dillon (2011, 2013) and Ching, Dillon, and Byrne (1998), maximizing the raw physical-audibility ANSI SII metric does not straightforwardly equate to maximized clinical benefit. While audibility is a powerful predictor of speech recognition in quiet, it explains progressively less variance as loss worsens. In listeners with severe loss, the magnitude of the hearing loss, reduced spectral resolution, and the number of cochlear dead regions account for roughly comparable shares of speech-in-quiet variance, with dead regions specifically blunting the benefit of restored high-frequency audibility (Vestergaard, 2003; Souza et al., 2018; Moualed, Humphries, & Ramsden, 2018). Empirically, aided speech recognition becomes insufficient in most ears

above ~ 80 dB HL regardless of how aggressively targets are met (Engler, Digeser, & Hoppe, 2026). Because Open-NL structurally optimizes for the raw SII metric, its highest-scoring targets drastically over-predict the usefulness of amplified regions in severely impaired ears. High amounts of prescribed gain may increase physical audibility on paper but severely degrade true perceptual clarity due to unmodeled suprathreshold distortion (Margolis et al., 2025).

To reconcile this limitation, Open-NL implements a bifurcated desensitization framework adapted from the empirically derived severe-loss desensitization formula of Johnson & Dillon (2011). Because the exact mathematical specification of the clinical formula—specifically its reliance on a piecewise $\max(0, \dots)$ gating mechanism—creates mathematical discontinuities (zero-gradients) that stall unconstrained numerical optimizers (Nelder-Mead), the internal optimizer evaluates targets against a continuous mathematical relaxation (`johnson2011_smoothed`). This smoothed heuristic safely anchors the optimization engine, permitting fluid gradient descent without trapping the algorithm in artificial local minima. Once the insertion gain targets are finalized, the resulting metrics are post-scored against the rigid clinical `johnson2011_complete` formula. As demonstrated in Table VI, this rigorous final penalty correctly mitigates the raw ANSI audibility for severe/profound profiles (such as aggressively penalizing A7 from 0.97 to 0.75), thereby structurally aligning the theoretical optimization outputs with realistic clinical expectations for cochlear dead regions. However, consistent with the overarching framework of the Open-NL testbed, it must be explicitly noted that the magnitude of this desensitization penalty is itself an unvalidated heuristic constant (now formally consolidated in Table III). Because the entire premise of this computational architecture is the interrogation of uncalibrated heuristics, this specific mathematical penalty

must be treated with the same scientific skepticism as the gain anchors it suppresses, pending systematic sensitivity mapping.

A. Proposal for Minimal Clinical Validation

While model-on-model evaluations computationally verify that the algorithm behaves as mathematically intended, they fundamentally cannot establish clinical utility. To bridge this gap, any minimal validation study of the Open-NL framework must be rigidly behaviorally anchored. First, as established by Valente et al. (2018), subjective outcomes and speech recognition scores are inextricably tied to precise acoustic matches in the patient’s ear canal. Any behavioral trial must utilize Real-Ear Measures (REM) to rigidly match the theoretical Open-NL targets; failing to achieve a verified real-ear match invalidates the subsequent behavioral data. Second, the study must establish a robust clinical control. Mueller (2005) demonstrated a lack of evidence supporting gain levels exceeding traditional NAL formulations for severe losses, and contemporary data from Engler, Digeser, and Hoppe (2026) explicitly favors conservative prescriptions (such as DSL v5.0) in the 50–80 dB HL range. Because Open-NL’s highest-SII targets prescribe mathematically derived gain that often exceeds these conservative baselines in severe profiles, a minimal validation study should structurally hypothesize that these extreme mathematical targets will trigger loudness rejection. By formally measuring this behavioral rejection boundary against a real-ear matched NAL-NL2 or DSL baseline, researchers can empirically constrain the necessary distortion penalties required to safely bind the optimizer in human subjects.

IV. SOFTWARE ARCHITECTURE AND THE INTERACTIVE DASHBOARD

A major objective of the **SII** package is translating complex acoustical mathematics into a usable format for both clinical researchers and audiological educators. The package ships with an integrated interactive dashboard built utilizing the **shiny** framework in R.

Rather than relying on web-based emulators or external servers, the dashboard is distributed natively within the CRAN package itself. By executing the `launch_app()` function, researchers can spin up a local instance of the application directly on their own machine. This native R execution ensures absolute numerical stability—avoiding the floating-point emulation inconsistencies inherent to WebAssembly (Wasm)—and guarantees that sensitive patient audiometric data never leaves the local computer. Furthermore, to reinforce its scope as a research framework rather than regulated diagnostic software, the application is explicitly flagged within its interface as a non-clinical academic tool, prohibiting its unapproved use for direct patient care.

The application provides a graphical user interface (GUI) where researchers can input standard audiograms, bone conduction thresholds, and LDLs. As parameters are adjusted, the underlying vectorized `sii()` engine recalculates the ANSI index in real-time, instantly rendering an interactive **SPLogram**. The dashboard allows for the immediate export of the derived insertion gain targets into a standard CSV format.

A. Code examples

The SII package utilizes a highly accessible, object-oriented API in R, facilitating the generation of reproducible figures for clinical publications via the command line.

Listing 1: Generating WDRC Targets

```
library(SII)

freqs <- c(250, 500, 1000, 2000, 3000, 4000, 6000, 8000)

thresholds <- c(20, 25, 40, 60, 65, 75, 80, 80)

# Calculate Open-NL for Average (65 dB)

sii_65 <- sii(speech = 65, threshold = thresholds,
              freq = freqs, prescription = "Open-NL")

# Extract the calculated insertion gain targets

sii_65$gain
```

V. CONCLUSION

Open-NL provides a transparent, customizable computational framework for predicting WDRC insertion gain targets and benchmarking them against established industry standards natively within R. By coupling an explicitly defined mathematical pipeline with an embedded computational loudness model, the SII package allows audiologists and researchers to freely simulate and evaluate the theoretical intelligibility-loudness trade-off of various prescriptive

heuristics without relying on opaque clinical fitting software. As the framework matures, it must eventually accommodate forthcoming prior art such as NAL-NL3 (Kitterick, Zakis, & Edwards, 2026). Because the NAL philosophy and its modular successor represent the established clinical standard, they are the natural comparators against which an open prescriptive substrate must be evaluated. It is critically important to emphasize that Open-NL is currently a computational toolkit designed exclusively for theoretical modeling and simulation. It has not undergone clinical validation and is strictly contraindicated for general clinical fitting. Given the inclusion of exploratory features like the severe-loss booster and dynamic MPO attenuation, applying these raw targets to human subjects carries an inherent risk of over-amplification or discomfort. Any future application of Open-NL targets to human subjects must occur exclusively within the strictly controlled context of Institutional Review Board (IRB) approved research studies, and mandates independent real-ear verification and rigorous loudness-discomfort safety checks.

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AUTHOR DECLARATIONS

Conflict of Interest

The author declares no conflicts of interest.

Ethics Approval

The author declares that no animal subjects or human participants were involved in the development, theoretical simulation, or mathematical validation presented in this research. Because Open-NL is distributed strictly as a theoretical computational toolkit, any future application of the algorithm to human participants by independent investigators requires separate Institutional Review Board (IRB) approval, informed consent, and rigorous audio-logical safety protocols. Furthermore, no identifiable data are embedded in the interactive Shiny application or the code repository.

DATA AVAILABILITY

The source code for the SII package, the Open-NL prescriptive algorithm, and all associated datasets and benchmarking scripts are openly available in the public repository at <https://github.com/r-gregmisc/SII> (v1.1.8; DOI: [DOI to be generated upon final repository release]; License: GPL-3.0).

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